

Asymptotics of assemblies of nested cycles

Exact counts and fixed order marked saddle expansions

Report 167

3 October 2026

Abstract

Let $P(z) = \prod_{m \geq 1} (1 + z^m/m)$ and $F(z, u) = \exp\{u(P(z) - 1)\}$. The labeled count $a_n = n![z^n]F(z, 1)$ is OEIS A308338, and the coefficient of u^k counts objects with k nonempty nested-cycle components, giving A392471. With $c = e^{-\gamma}$, we prove

$$a_n \sim \frac{e^{c(1/2 - \log 2) - 1} c^{1/4 - c/2}}{2\sqrt{\pi}} n! n^{c/2 - 3/4} e^{2\sqrt{cn}}.$$

More generally, every fixed order has a complex-uniform expansion in $n^{-1/2}$ with polynomial coefficients in $\log n$; we give the first two corrections explicitly. A whole-circle estimate controls all secondary arcs, and a Gaussian Taylor lemma gives the stated sharp logarithmic remainder without crossing a natural boundary.

For the component count, holomorphic normalization justifies differentiation and gives explicit logarithmic and constant corrections to the mean and variance. In particular the mean divided by $\sqrt{n}$ tends to $e^{-\gamma/2}$. We recover the established Gaussian regime with variance equivalent $\sqrt{cn}/2$ and prove a rounding-safe asymptotic inverse for the integer count sequence.

The component asymptotics and generic expansive-assembly limit laws are prior results. The contribution developed here is the explicit sequence-specific marked expansion and its refinements. The companion contains bounded exact integer and symbolic checks. No worldwide priority claim, complete beyond-all-orders transseries, or certified numerical asymptotic error bound is asserted.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 167 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a308338-nested-cycle-assemblies** (provenance, the sources as the write read them, relation to the repository, the collected non-claims and the reading conventions in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 167” and its PDF author field is empty. *What rests on what.* Everything analytic rests on the component estimate (1.4), credited to Greene and Knuth through Flajolet, Fusy, Gourdon, Panario and Pouyanne and cited, not reproved. Given it, the text proves the fixed-order expansion (Theorem 2.1), the moment expansions (Theorem 7.1), the Gaussian limit (Corollary 7.2, a known regime re-proved with this normalization) and the two-ceiling inverse (Theorem 8.1); no constant or onset is made effective and every decimal is uncertified. *Two statements of the literature are put on record, with proofs.* (i) Riedel’s January 2026 conjecture $E[X] \sim N\sqrt{n}$ for the number of components is *proved* by (7.8), with $N = e^{-\gamma/2} = 0.7493\dots$; this excludes the alternative he raised, a power n^M with M “close to but not equal to $1/2$ ”, and his numerical $N \approx 0.7719351399$ is not the constant but agrees to nine decimals with the finite ratio $\mathbb{E}K_{4500}/\sqrt{4500}$ (Remark 7.3). The source quotes the estimate without saying that it settles the conjecture. (ii) The general- C covariance display of Erlihson and Granovsky (preprint (4.54), published (4.18), both read) is negative at $p = 1$, $C = 3$, $u = 1$ and contradicts Corollary 7.2 in this report’s own model; the write gives the corrected display and locates the dropped factor C (Remark 10.1). The source had already found the negative value. The write also adds Remark 8.2 (the inversion against the transseries volume), notes on the OEIS entries as of 6 October 2026, and one question (Section 11). Nothing was submitted to the OEIS or sent to any author.

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Scope and reading guide. Sections 1–2 specify the object and results. Sections 3–5 prove the fixed-order coefficient theorem. Sections 6–8 give corrections, probability laws, and inversion. Section 9 separates exact verification from numerical diagnostics. Section 10 records source coverage and limits. The report is self-contained after the explicitly credited component coefficient estimate (1.4).

1 Exact objects and established input

A labeled cycle on an m -element set has $(m-1)!$ possible cyclic orders and exponential generating function z^m/m . Choosing at most one cycle of each length therefore gives

$$P(z) = \prod_{m \geq 1} \left(1 + \frac{z^m}{m}\right) = \sum_{m \geq 0} B_m \frac{z^m}{m!}, \quad p_m = \frac{B_m}{m!}. \quad (1.1)$$

The integers B_m form A007838. A nonempty collection of cycles of distinct lengths has a unique decreasing-length nesting order. It is one nested component in Riedel's interpretation; there is no extra ordering or geometric-embedding choice. A set of such nonempty labeled components has marked exponential generating function

$$F(z, u) = \exp\{u(P(z) - 1)\}, \quad f_n(u) = [z^n]F(z, u). \quad (1.2)$$

Thus

$$T(n, k) = \frac{n!}{k!} [z^n](P(z) - 1)^k, \quad a_n = n! f_n(1) = \sum_{k=0}^n T(n, k). \quad (1.3)$$

The a_n are A308338, with $a_0 = 1$, and the published triangle A392471 starts at $n = 1$, $1 \leq k \leq n$. We use the natural empty extension $T(0, 0) = 1$ and $T(n, 0) = 0$ for $n > 0$ only for recurrences. In particular $T(n, 1) = B_n$ for $n \geq 1$.

Ilya Gutkovskiy entered A308338 and its defining EGF on 20 May 2019 [2]. Riedel's January 2026 note supplies the nested-cycle interpretation, the marked triangle, and recurrences [4, 3]. The component EGF is much older. The prior estimate we use is

$$p_m = c + \frac{c}{m} + O\left(\frac{\log m}{m^2}\right), \quad c = e^{-\gamma}, \quad m \rightarrow \infty, \quad (1.4)$$

where γ is Euler's constant. Flajolet, Fusy, Gourdon, Panario, and Pouyanne [5, Section 3, equation (22)] attribute this estimate to Greene and Knuth. Their Proposition 1 gives the full component coefficient expansion, including periodic terms from roots of unity. We use (1.4) as a credited theorem; their theorem for P is not itself a coefficient theorem for the outer exponential F .

For a component spectrum $(j_1, \dots, j_n)$ with $\sum m j_m = n$, the number of objects is

$$n! \prod_{m=1}^n \frac{p_m^{j_m}}{j_m!}.$$

After division by a_n , this is the Gibbs assembly law with weights $p_m \sim c$, so its expansive exponent is 1. The weights are p_m , not B_m or p_m/m . The square-root component law and normal fluctuations lie within established expansive-assembly theory. Erlihson and Granovsky [6, Corollary 4.5] explicitly attribute the total-component CLT to their 2004 work [7]. We derive the precise normalization and lower-order terms directly below.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 167 of the session bundle that ProveIt's docs/incoming intake processed as batch 108: the archive `Nested_Cycle_Assemblies_Asymptotics_and_Inverses_Source.zip` (938,156 bytes, SHA-256 `ef59faff7aba...eaeabc0b63`, 21 files at the archive root and in its `companion/` and `data/` folders), arrival commit `60f54ea06`, placed in `602e5bd0f`.

The text is the delivered `Report167.tex` (908 lines, dated 3 October 2026), shipped as `article.tex`. The delivered 15-page PDF and the checksum manifest `SHA256SUMS` (20 entries, verified at placement and again at the write) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data; the word “private” occurs only in the delivered README’s “private fresh format” (a `TeX` format file). It is a single-source report, so no merge choice was made. The write prefixed the 62 delivered labels with `ncy:` and updated the 60 references to them; it added this subsection, the notes marked [write], Remarks 7.3, 8.2 and 10.1, labels on the items of Section 11 and one further item there. No statement, proof or number of the source was changed: the added remarks are the last statements of their sections, the added displays are unnumbered, and the added item follows the last delivered one, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A308338 [2], live revision #21 (5 October 2026); the source froze revision #17 (15 January 2026). Revisions #18–#21, by Alois P. Heinz on 5 October 2026, add a Maple program and a b-file for $0 \leq n \leq 445$; the 22 data terms are unchanged. Riedel’s comment identifying the terms as nested cycle partitions and his formula $a(n) = \sum_k T(n, k)$ date from 14 January 2026. The entry states no asymptotic and no conjecture.
- OEIS A392471 [3], live revision #42 (5 October 2026); the source froze revision #37 (25 January 2026). On 5 October 2026 Heinz inserted the column $k = 0$ (the entry’s `%E` line reads “Column $k=0$ inserted by Alois P. Heinz, Oct 05 2026”), so the entry now has offset 0 and rows from $n = 0$, and he added a Maple program and a b-file of rows $0 \leq n \leq 150$. The delivered sentence of Section 1 that the published triangle “starts at $n = 1$, $1 \leq k \leq n$ ” describes revision #37; the entry now uses the same extension $T(0, 0) = 1$, $T(n, 0) = 0$ ($n > 0$) as this report. Its link to Riedel’s original note is titled “Nested Cycle Partitions: definitions and basic recurrences”; the PDF behind it is titled “Nested Cycle Partitions: a conjecture”. The entry itself contains no conjecture.
- OEIS A007838 [1], revision #74 (13 March 2026), unchanged; its formula section states (1.4) in the form $a(n) \sim n!(e^{-g} + e^{-g}/n + O((\log n)/n^2))$ and its reference section cites Greene and Knuth.
- The three b-files agree with `data/exact_data_100.json` for $0 \leq n \leq 100$ (A308338, A007838) and for all rows $0 \leq n \leq 100$ of the triangle (the write’s comparison).
- Riedel’s note [4] in both versions: the revised one (author-hosted, page footers dated 24.01.26) and the original linked from A392471 (footers dated 13.01.26), six pages each; the paragraph “The conjecture” on p. 3 is the same in both (Remark 7.3).
- Erlihson and Granovsky [6]: the preprint arXiv:math/0507343v3 (19 June 2007), Section 4 from (4.37) to (4.55) and Section 5 around (5.121), and the published text (numdam), pp. 925–927 and 936–938 (Remark 10.1).
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 8.2.
- Not read by the write: the hybrid paper [5] (the source read its Section 3 and Proposition 1), the Greene–Knuth book and the 2004 paper of Erlihson and Granovsky [7] (the source read neither). `data/SOURCE_PROVENANCE.json` records what the source inspected.

What was checked. The batch-108 dossier read the manuscript in full and found no false claim. It checked (1.4) at $m = 100$ ($p_{100} = 0.56678641$ against $c + c/100 = 0.56707408$, a difference within $\log m/m^2$), the counts, the exact mean and the exact variance at $n = 100$ against (2.3) and Theorem 7.1, read Riedel’s conjecture paragraph, verified the literal substitution in the Erlihson–Granovsky preprint, and replayed the companion on copies (all four frozen outputs reproduced). At the write the companion was rerun from a fresh extraction of the archive and from the shipped files (note at the end of Section 9), and $f_n(1)$, $\mathbb{E}K_n$ and $\text{Var } K_n$ were evaluated in double precision for $n \leq 40000$ by recurrences with positive terms only (p_m from the cycle product of (9.1) divided by $m!$, $f_n(1)$ from the second recurrence of (9.3), and the moments

from $[z^n]F(z, 1)(P(z) - 1)$ and $[z^n]F(z, 1)(P(z) - 1)^2$, agreeing with the exact rational values at $n = 100$ ($\mathbb{E}K_{100} = 8.219543$, $\text{Var } K_{100} = 4.016577$). The residuals are consistent with Theorem 2.1 at $J = 1$ and with Theorem 7.1 including M_1 and V_1 : at $n = 40000$, $\sqrt{n}(a_n/(Cn!n^{c/2-3/4}e^{2\sqrt{cn}}) - 1) = 9.682$ against $R_1(\log n; 1) = 9.344$, $\sqrt{n}$ times the residual of (7.2) is 12.92 against $M_1(\log n) = 12.69$, and $\sqrt{n}$ times the residual of (7.3) is 16.97 against $V_1(\log n) = 16.53$; at $n = 1000$ the same pairs are 5.43/4.71, 6.74/6.13 and 8.38/7.27, so the gaps shrink as the next order predicts. These are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository named A308338, A392471 or A007838, nested cycle partitions or Riedel’s note, and none does outside this report and the collection catalogue. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The neighbouring reports `a064856-stirling-catalan-transforms` (Part II: cycle-weighted permutation mixtures) and `a301746-divisor-weighted-asymptotics`, `a022629-distinct-partition-norms`, `a271619-strict-twice-partitions` (Gibbs partitions after Granovsky, Stark and Erlihson) treat other models; none uses the Erlihson–Granovsky display corrected in Remark 10.1, and none needs a reciprocal note. The inversion of Section 8 is compared with the transseries volume in Remark 8.2.

[write] What is not claimed, collected from the source. The expansion of Theorem 2.1 holds at every fixed order only: no convergence, optimal truncation, Stokes phenomenon or complete beyond-all-orders description is claimed, and constants need not be uniform when the order grows. The decimal value of C is an uncertified floating evaluation; no interval enclosure of C is supplied, and no implied constant or onset (in particular no certified M_J or threshold in Theorem 8.1) is made effective. The calculator (6.3) is formal: its contour is not identified with a Bessel contour and no convergence is claimed. The square-root law, the exact limit (7.8) and the normal fluctuations belong to the established expansive-assembly regime; Corollary 7.2 is “not a new universality assertion”, and no local limit theorem or distribution distance bound is inferred. The finite and symbolic checks prove no asymptotic statement; the B , B_3 checks verify the arithmetic pieces, not the infinite-tail summations. Source coverage is bounded: no worldwide priority is claimed, “No claim is made that no other work contains overlapping results”, the Greene–Knuth book and the 2004 paper were not inspected, and no OEIS entry was amended. Byte identity of rebuilds is toolchain-specific. The write adds: the table and the identification of $n = 4500$ in Remark 7.3 are floating evaluations (checked against exact and 256-bit fixed-point values) and an inference about how Riedel’s number arose, not a statement of his; the corrected display of Remark 10.1 is derived from the conditioning computation, not from a re-audit of the whole proof of Erlihson and Granovsky.

Reading conventions (letters the source reuses, and letters of the two outside sources discussed by the write, with the tempting false readings; no symbol of the source was renamed; ϱ and $M_\varrho(n)$ are the write's).

Symbol	Meaning here (where)	Not to be read as
B_m, p_m	component counts (A007838) and $B_m/m!$	the constants B, B_3 of (3.4); the calculator $\mathcal{B}_\beta$; Erlihson–Granovsky's b_r and exponent p
c, A, D, E	$e^{-\gamma}$; constants of (2.1), (3.6)	OEIS A-numbers; $\mathbb{E}$; the residual $E_{n,J}$
$C(u), C$	amplitude (2.1), (2.3)	$C_0, C_{j,k,m}$ (proof constants); Erlihson–Granovsky's C in $a_k \sim Ck^{p-1}$
$u, v = cu$	marking variable; cu	Erlihson–Granovsky's scaled size u ; v of <code>p0:prop:factorial-core</code>
L	$\log n$ in $R_j(L; u), M_1(L), V_1(L)$; $\log y$ in Section 8	$L_n(u)$ (leading term); the target L of <code>p0:prop:factorial-core</code>
$Q(X), Q_j, X_n$	polynomials of (3.6), (5.2); contour bound	Riedel's $Q(z, u) = F(z, u)$ and $Q_k(z)$; Riedel's $X = K_n$
K_n, K, K_0	component count; constant in the proof of Theorem 8.1; $\log(C\sqrt{2\pi})$	
$H_{j,k}, H_j$	Taylor factor (5.5); harmonic numbers	
ℓ, ℓ_c, ℓ_r	$\log s$; $(\log c - L)/2$; coefficients of $\log P$ (Section 9)	
β	$c/2 - 3/4$ (Section 8)	the free exponent in $\mathcal{B}_\beta$
$\kappa, \kappa', h, d, \delta$	loss constants, window $a^{3/2} \log n$ (Section 4); bound and $n^{-1/4}$ in Lemma 5.1	the volume's $\kappa_{\text{vol}}, d_{\text{vol}}, h$; Riedel's κ_n ; Erlihson–Granovsky's $\delta = 1/r_N$
p, q	$2J + 2, (J + 1)/2$ (Section 8); $q = Q(\ell_c)$ in (7.6)	p_m ; Erlihson–Granovsky's exponent p and number q of strata
$N(y), M_J, M_1$	first passage; constant of Theorem 8.1; mean correction	Riedel's constant N and exponent M ; Erlihson–Granovsky's size N ; the write's $M_\varrho(n)$
$T(n, k), t$	A392471 (with $T(n, 0)$); $z = e^{-t}$	

2 The marked coefficient theorem

Fix a sufficiently small open complex disc about $u = 1$. Choose $\log(cu)$ and $\sqrt{cu}$ to be real on positive real u , and set

$$A = \frac{1}{2} - \log 2, \quad C(u) = \frac{e^{u(cA-1)}(cu)^{1/4-cu/2}}{2\sqrt{\pi}}. \quad (2.1)$$

All statements about fixed order mean that the order is fixed as $n \rightarrow \infty$; constants need not be uniform when that order grows.

Theorem 2.1 (Uniform fixed-order expansion). *For every fixed integer $J \geq 0$ there are polynomials*

$R_j(L; u)$ of degree at most $2j$ in L , with coefficients holomorphic near 1, such that

$$f_n(u) = C(u)n^{cu/2-3/4}e^{2\sqrt{cn}} \left\{ 1 + \sum_{j=1}^J n^{-j/2} R_j(\log n; u) + E_{n,J}(u) \right\}, \quad (2.2)$$

$$E_{n,J}(u) = O\left(n^{-(J+1)/2}(1 + \log n)^{2J+2}\right).$$

The estimate is uniform on every sufficiently small fixed closed disc about 1. The exact normalized residual $E_{n,J}$ is holomorphic on an open disc containing any smaller disc on which derivatives are taken, and each fixed derivative obeys the same order bound there.

Taking $u = 1$, $J = 0$ gives the equivalent

$$a_n \sim Cn!n^{c/2-3/4}e^{2\sqrt{cn}}, \quad C = \frac{e^{c(1/2-\log 2)-1}c^{1/4-c/2}}{2\sqrt{\pi}}. \quad (2.3)$$

For orientation only, $C \approx 0.0947779253874791584250013230745944160$; this is an uncertified floating evaluation of the exact expression. The logarithmic term of P supplies the power $n^{c/2}$ in (2.3). Discarding that term would give an incorrect equivalent.

The theorem is an all-fixed-orders algebraic expansion with logarithmic polynomials. It makes no claim about convergence, optimal truncation, Stokes phenomena, or a complete description of terms beyond all orders.

3 Local component expansion in a right half plane

We first prove the needed local expansion without continuing P through its natural boundary. For $\Re t > 0$, isolating the factor $m = 1$ gives the absolutely convergent identity

$$\begin{aligned} \log P(e^{-t}) &= \log(1 + e^{-t}) + \text{Li}_1(e^{-t}) - e^{-t} \\ &\quad + \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} \{\text{Li}_k(e^{-kt}) - e^{-kt}\}. \end{aligned} \quad (3.1)$$

Fix a desired order H and choose an integer $K > H + 2$. The tail $k > K$ can be written as a double sum over $k > K$, $m \geq 2$. For $0 \leq h \leq H + 1$, its h th derivative in t is bounded absolutely and uniformly on the closed right half plane by

$$\sum_{k > K} k^{h-1} \sum_{m \geq 2} m^{h-k} < \infty. \quad (3.2)$$

Indeed $k - h > 1$ and the inner sum decreases exponentially as k grows. Termwise differentiation up to that order is consequently valid. The integral remainder along the segment from 0 to t gives a Taylor polynomial for the tail with error $O(|t|^{H+1})$, uniformly in the closed right half plane near 0.

Each retained polylogarithm has the classical finite-order expansion

$$\text{Li}_k(e^{-x}) = \frac{(-x)^{k-1}}{(k-1)!} (H_{k-1} - \log x) + \sum_{\substack{r \geq 0 \\ r \neq k-1}} \zeta(k-r) \frac{(-x)^r}{r!}, \quad (3.3)$$

where $H_j = \sum_{m=1}^j 1/m$ and $H_0 = 0$. For a fixed finite set of k , truncations and their remainders are uniform in every closed subsector of $\Re t > 0$. Combining them with (3.2) gives each fixed order

of $\log P(e^{-t})$. At every positive power t^r , the coefficient is affine in $\log t$. Its logarithmic coefficient comes only from $k = r + 1$ in (3.1) and is

$$-\frac{(r+1)^{r-1}}{r!} \quad (r \geq 1).$$

The constant after removing $-\log t$ is $-\gamma$, by the convergent Euler product with the harmonic term subtracted. Explicitly,

$$\begin{aligned} \log P(e^{-t}) = & -\log t - \gamma + t(-\log t + A) + t^2 \left(-\frac{3}{2} \log t + B \right) \\ & + t^3 \left(-\frac{8}{3} \log t + B_3 \right) + O(|t|^4(1 + |\log t|)), \end{aligned} \quad (3.4)$$

$$B = \frac{41}{24} - \frac{\zeta(2)}{2} - \frac{3}{2} \log 3, \quad B_3 = \frac{49}{12} - \frac{\pi^2}{4} + \frac{\zeta(3)}{3} - \frac{8}{3} \log 4.$$

For completeness, the t^2 constant contributions from the isolated prefactor, $k = 2$, $k = 3$, and $k \geq 4$ are, respectively,

$$-\frac{5}{12}, \quad \frac{3}{2}, \quad \frac{3}{4} - \frac{3}{2} \log 3, \quad -\frac{\zeta(2)}{2} - \frac{1}{8}.$$

The last follows by summing a rational series:

$$\begin{aligned} \frac{1}{2} \sum_{k \geq 4} (-1)^{k+1} k(\zeta(k-2) - 1) &= -\frac{1}{2} \sum_{m \geq 2} \frac{4m+3}{m(m+1)^2} \\ &= -\frac{\zeta(2)}{2} - \frac{1}{8}. \end{aligned}$$

For B_3 , the corresponding contributions (now separating $k = 4$) are $1/6$, $-13/18$, $9/4$, $20/9 - (8/3) \log 4$, and

$$-\frac{1}{6} \sum_{m \geq 2} \frac{25m^2 + 39m + 16}{m(m+1)^3} = \frac{1}{6} - \frac{3}{2} \zeta(2) + \frac{1}{3} \zeta(3).$$

These calculations supply the constants rather than attributing them to a short displayed formula in the component source.

Exponentiation to any prescribed fixed order yields

$$P(e^{-t}) = \frac{c}{t} + c(-\log t + A) + \sum_{j=1}^M t^j U_j(\log t) + O(|t|^{M+1}(1 + |\log t|)^{M+2}), \quad (3.5)$$

where $\deg U_j \leq j + 1$. To obtain the stated error, first expand $\log P$ through degree $M + 1$, with error $O(|t|^{M+2}(1 + |\log t|))$, and then exponentiate. The largest omitted logarithmic degree comes from $M + 2$ copies of $t \log t$. Define

$$\begin{aligned} D &= \log 2 - 2, \quad E = B + A^2/2, \quad Q(X) = X^2/2 + DX + E, \\ Z(X) &= \frac{(-X + A)^3}{6} + (-X + A) \left(-\frac{3}{2}X + B \right) - \frac{8}{3}X + B_3. \end{aligned} \quad (3.6)$$

Then $U_1 = cQ$ and $U_2 = cZ$. All these are sector expansions inside the right half plane. No analytic continuation of P across $|z| = 1$ has been used.

4 The complete circle and complex saddle

The global step comes from the prior coefficient estimate rather than from a claim that the positive real singularity is the only singularity. From (1.4), absolute summation of $|p_m - c|e^{-am}$, including the bounded $m = 0$ adjustment, gives uniformly for $0 < a < 1$, $|y| \leq \pi$,

$$P(e^{-a-iy}) = \frac{c}{1 - e^{-a-iy}} + O(1 + \log(1/a)) = \frac{c}{a + iy} + O(1 + \log(1/a)). \quad (4.1)$$

For the second equality, $(1 - e^{-t})^{-1} - t^{-1}$ is bounded on $0 \leq \Re t \leq 1$, $|\Im t| \leq \pi$ after filling its removable value at 0. Thus (4.1) includes every secondary root-of-unity arc.

Put $v = cu$ and $s = \sqrt{v/n} = a + ib$, where $a > 0$. Coefficient extraction on $|z| = e^{-a}$, with $z = e^{-t}$, gives

$$f_n(u) = \frac{1}{2\pi i} \int_{a-i\pi}^{a+i\pi} \exp\{u(P(e^{-t}) - 1) + nt\} dt. \quad (4.2)$$

The sign from $dz/z = -dt$ is canceled by reversal of the circle parameter. The principal phase is $\Phi(t) = v/t + nt$, with saddle s . Since $v = ns^2$, direct algebra gives the exact loss identity

$$\Re \Phi(a + iy) = 2na - na \frac{(y - b)^2}{a^2 + y^2}. \quad (4.3)$$

Uniformly on the small marking disc, $a \asymp n^{-1/2}$ and $|b|/a$ is bounded. Take $h = a^{3/2} \log n$. Outside $|y - b| < h$, the loss in (4.3) is at least $\kappa(\log n)^2$ for a positive constant κ . To see this, when $|y| \leq C_0 a$ the denominator is $O(a^2)$ and $nah^2/a^2 \asymp (\log n)^2$; when $|y| > C_0 a$ with C_0 sufficiently large, the fraction is bounded below, and the loss is of order $\sqrt{n}$.

The error from (4.1) costs only $O(\log n)$ in the real exponent, with u bounded. The entire off-saddle contribution is therefore

$$O\left(e^{2na - \kappa'(\log n)^2}\right) \quad (4.4)$$

for a uniform $\kappa' > 0$. The modulus of the leading term in (2.2) is e^{2na} times a quantity bounded above and below by fixed powers of n . Hence (4.4), relatively to that term, is smaller than every fixed inverse power of n . This proves the needed suppression for the whole circle, including the natural-boundary points, rather than relying only on positivity or aperiodicity.

5 The local Gaussian expansion and its remainder

On $|t - s| \leq h$, t remains in a fixed right-half-plane subsector and $|t| \asymp n^{-1/2}$. Equation (3.5) gives, for any fixed J ,

$$\begin{aligned} e^{u(P(e^{-t})-1)} &= e^{u(cA-1)} t^{-v} e^{v/t} \\ &\times \left\{ \sum_{j=0}^J t^j Q_j(\log t; u) + O(|t|^{J+1}(1 + |\log t|)^{2J+2}) \right\}. \end{aligned} \quad (5.1)$$

Here $Q_0 = 1$, and all Q_j are defined by the formal identity

$$\exp \left(u \sum_{j \geq 1} t^j U_j(X) \right) = \sum_{j \geq 0} t^j Q_j(X; u). \quad (5.2)$$

A product of r positive-degree factors with total t -degree j has logarithmic degree at most $j+r \leq 2j$; therefore $\deg_X Q_j \leq 2j$. Truncating the ordinary exponential gives the error in (5.1), since $t(1+|\log t|)^2 \rightarrow 0$ on the segment. The component error in (3.5) is no larger. In particular,

$$Q_1 = vQ, \quad Q_2 = vZ + \frac{v^2}{2}Q^2. \quad (5.3)$$

The next lemma records the integrated Taylor remainder carefully; replacing powers of the integration variable by endpoint bounds would lose unnecessary powers of $\log n$.

Lemma 5.1 (Uniform Gaussian Taylor remainder). *Set $\delta = n^{-1/4}$, $\lambda = \sqrt{vn}$, and use the actual vertical contour parametrization*

$$t = s + in^{-3/4}x = s(1 + i\delta x/\sqrt{v}), \quad |x| \leq X_n = (\Re\sqrt{v})^{3/2} \log n.$$

For fixed nonnegative integers j, k, S , the integral of the monomial $t^{j-v}(\log t)^k$ times $e^{nt+v/t}$ has an expansion, after removing its common leading scale, in powers $n^{-r/2}$, $0 \leq r \leq S$. Its relative-to-that-scale remainder is $O(n^{-(S+1)/2}(1+|\log s|)^k)$, uniformly in u on a sufficiently small fixed closed disc. Each coefficient is polynomial in $\log s$ of degree at most k and holomorphic in u .

Proof. The exact phase, with no truncated exponential yet, is

$$nt + \frac{v}{t} = 2\lambda - \frac{x^2}{\sqrt{v}} + \frac{i\delta x^3}{v(1 + i\delta x/\sqrt{v})}. \quad (5.4)$$

Write $\ell = \log s = (\log v - \log n)/2$ and remove s^{j-v} from the monomial. The residual analytic factor multiplying the Gaussian is

$$\begin{aligned} H_{j,k}(\delta, x; v, \ell) &= (1 + i\delta x/\sqrt{v})^{j-v} [\ell + \log(1 + i\delta x/\sqrt{v})]^k \\ &\quad \times \exp\left(\frac{i\delta x^3}{v(1 + i\delta x/\sqrt{v})}\right). \end{aligned} \quad (5.5)$$

In a Taylor expansion in δ , hold ℓ fixed; substitute its stated value afterwards. Let $d > 0$ be a uniform lower bound for $\Re(v^{-1/2})$ on the marking disc. For every fixed derivative order m and all sufficiently large n , one has, for $0 \leq \xi \leq \delta$ and $|x| \leq X_n$,

$$|\partial_\delta^m H_{j,k}(\xi, x; v, \ell)| \leq C_{j,k,m}(1 + |\ell|)^k P_{j,k,m}(|x|)e^{dx^2/2}, \quad (5.6)$$

with a fixed polynomial $P_{j,k,m}$ having nonnegative coefficients. Indeed, $|\xi x/\sqrt{v}| \leq 1/2$ eventually. Differentiation of the rational exponent and logarithm gives fixed polynomial bounds in $|x|$, with denominators uniformly separated from zero. The exponential modulus is at most $\exp(C\delta|x|^3)$, and $C\delta X_n \leq d/2$ eventually. The complex power and its fixed number of derivatives are bounded in the same way by polynomials. Finite product and chain rules prove (5.6).

Expand (5.5) through degree $2S+1$. The integral Taylor remainder, multiplied by $e^{-x^2/\sqrt{v}}$ and integrated, is at most

$$C\delta^{2S+2}(1 + |\ell|)^k \int_{\mathbb{R}} P_{j,k,2S+2}(|x|)e^{-dx^2/2} dx = O(n^{-(S+1)/2}(1 + |\ell|)^k). \quad (5.7)$$

This uses Gaussian moments, not the bound $|x| \leq X_n$ for the polynomial factor, so it introduces no extra powers of $\log n$. The identity $H_{j,k}(-\delta, -x) = H_{j,k}(\delta, x)$ implies that the coefficient of δ^m

has parity m in x . All odd coefficients integrate to zero on the symmetric centered segment, also for complex u because its center is the actual saddle s . The remaining even coefficients can be integrated over $\mathbb{R}$: the tails beyond $X_n \asymp \log n$ are smaller than every fixed inverse power of n .

The moments used are

$$\int_{\mathbb{R}} x^{2r} e^{-x^2/\sqrt{v}} dx = \sqrt{\pi} \frac{(2r-1)!!}{2^r} v^{(2r+1)/4}, \quad \int_{\mathbb{R}} x^{2r+1} e^{-x^2/\sqrt{v}} dx = 0, \quad (5.8)$$

with $(-1)!! = 1$. They hold first for real positive v by substitution and then throughout the disc by holomorphy of dominated Gaussian integrals. This use of continuation concerns the Gaussian integral only. Polynomial dependence on ℓ and holomorphic dependence on u are preserved by the finite Taylor and moment operations. $\square$

Proof of Theorem 2.1. For a monomial in Q_j , apply Lemma 5.1 with $S = J - j$. Its s^j contributes $n^{-j/2}$, and its logarithmic degree is at most $2j$. Its total relative error is therefore

$$O(n^{-(J+1)/2} (1 + \log n)^{2j}).$$

The remainder in (5.1), integrated absolutely with the local Gaussian, contributes $O(n^{-(J+1)/2} (1 + \log n)^{2J+2})$ relative to the leading scale. Equation (4.3), or (5.4) with its small correction, supplies a fixed Gaussian majorant; t^{-v}/s^{-v} is bounded on the segment. Equation (4.4) disposes of the rest of the circle.

The Jacobian and leading Gaussian integral give precisely

$$\frac{n^{-3/4}}{2\pi} e^{u(cA-1)} s^{-v} e^{2\lambda} \sqrt{\pi} v^{1/4} = C(u) n^{v/2-3/4} e^{2\sqrt{vn}}.$$

Every even correction coming from Q_j has logarithmic degree at most $2j$. At total order $n^{-r/2}$, $j \leq r$, proving the degree bound $2r$ and the coefficient holomorphy.

Finally, let $L_n(u) = C(u) n^{cu/2-3/4} e^{2\sqrt{cun}}$. It is holomorphic and nonzero on a small fixed open disc. Since $f_n(u)$ is a polynomial, the exact expression

$$E_{n,J}(u) = \frac{f_n(u)}{L_n(u)} - 1 - \sum_{j=1}^J n^{-j/2} R_j(\log n; u) \quad (5.9)$$

is holomorphic there. The proof gives its bound on every sufficiently small closed disc. Choose two concentric discs whose closures are inside the open disc; Cauchy's derivative estimates transfer the bound on the larger disc to all fixed derivatives on the smaller disc, without changing its powers of n or $\log n$. The contour radius $a = \Re \sqrt{cu/n}$ need not vary holomorphically: it is a device for proving a bound on the already holomorphic expression (5.9). $\square$

6 Explicit corrections and a fixed order calculator

Retain $v = cu$, $\ell = (\log v - L)/2$ and Q, Z, D from (3.6). The first two correction polynomials are

$$R_1(L; u) = v^{3/2} Q(\ell) - \frac{4(v-1)^2 - 1}{16\sqrt{v}} \quad (6.1)$$

and

$$\begin{aligned}
R_2(L; u) = & \frac{[4(v-1)^2 - 1][4(v-1)^2 - 9]}{512v} \\
& + v \left\{ \frac{1 - 4(v-2)^2}{16} Q(\ell) + \frac{v-2}{2}(\ell + D) - \frac{1}{4} \right\} \\
& + v^2 Z(\ell) + \frac{v^3}{2} Q(\ell)^2.
\end{aligned} \tag{6.2}$$

In particular the coefficient of L^2 in R_1 is $v^{3/2}/8$.

Here is a direct Gaussian check of the first correction. The unperturbed amplitude t^{-v} has order- δ^2 integrand

$$-\frac{v+1}{2}x^2 + \frac{v+1}{v^{3/2}}x^4 - \frac{1}{2v^2}x^6.$$

The normalized Gaussian moments are $\sqrt{v}/2$, $3v/4$, and $15v^{3/2}/8$, giving $-[4(v-1)^2 - 1]/(16\sqrt{v})$. The tQ_1 term contributes $v^{3/2}Q(\ell)$, proving (6.1).

For clarity, the four parts of (6.2) have separate origins. The unperturbed fourth-order term gives its first line. The order- δ^2 correction to tQ_1 gives its second line. The leading Gaussian integral of t^2Q_2 gives its last line, by (5.3). Each follows from (5.5) and the exact moments (5.8); the companion repeats these identities symbolically. This makes R_2 a finite algebraic consequence of the same local calculation, rather than an additional asymptotic assumption.

For general fixed order, equations (3.1), (5.2), and (5.5) are an algorithm using only finitely many coefficients and Gaussian moments. An equivalent compact formal calculator for a monomial $t^{-\beta}$ is

$$\begin{aligned}
\mathcal{B}_\beta(v, n) = & \left(\frac{n}{v}\right)^{(\beta-1)/2} \frac{e^{2\sqrt{vn}}}{\sqrt{4\pi\sqrt{vn}}} \\
& \times \sum_{r \geq 0} \frac{(-1)^r \prod_{l=1}^r [4(\beta-1)^2 - (2l-1)^2]}{r!(16\sqrt{vn})^r}.
\end{aligned} \tag{6.3}$$

For $t^{j-v}(\log t)^k$, replace the monomial by $(-\partial_\beta)^k \mathcal{B}_\beta(v, n)$ at $\beta = v - j$, holding v fixed, multiply by $e^{u(cA-1)}$, and retain terms with $j + r \leq J$. Equation (6.3) is the dominant-saddle formal series familiar from the modified Bessel function. Its use here is a coefficient calculator: the original full contour has not been identified with a Bessel contour, and no convergence of the formal series is claimed. The remainder proof is the Gaussian argument of Section 5.

7 Refined component moments and the Gaussian limit

Let K_n be the number of nested components in an object chosen uniformly from the a_n labeled objects. Its probability generating function is

$$\mathbb{E}u^{K_n} = \frac{f_n(u)}{f_n(1)}.$$

Since $f_n/L_n \rightarrow 1$ uniformly near $u = 1$, the ratio is nonzero on a fixed smaller disc for all sufficiently large n and has the analytic logarithm tending to zero there. The theorem and finite logarithmic expansion imply

$$\log f_n(u) = \log L_n(u) + n^{-1/2}R_1(\log n; u) + O(n^{-1}(1 + \log n)^4), \tag{7.1}$$

with the same order for any fixed number of derivatives on a smaller disc. This follows by applying Cauchy's estimates to the exact normalized residual before differentiation. Thus no real-axis asymptotic is being differentiated without justification.

Theorem 7.1 (Mean and variance). *As $n \rightarrow \infty$,*

$$\mathbb{E}K_n = \sqrt{cn} + \frac{c}{2} \log n + m_0 + O(n^{-1/2}(1 + \log n)^2), \quad (7.2)$$

$$m_0 = -\frac{3}{4} - c \log 2 + \frac{c\gamma}{2},$$

$$\text{Var } K_n = \frac{\sqrt{cn}}{2} + \frac{c}{2} \log n + v_0 + O(n^{-1/2}(1 + \log n)^2), \quad (7.3)$$

$$v_0 = -1 - c \left(\log 2 + \frac{1}{2} - \frac{\gamma}{2} \right).$$

The next corrections are respectively $n^{-1/2}M_1(\log n)$ and $n^{-1/2}V_1(\log n)$, after which both errors are $O(n^{-1}(1 + \log n)^4)$, where

$$M_1(L) = c\partial_v \mathcal{R}_1(L, v)|_{v=c}, \quad V_1(L) = (c\partial_v + c^2\partial_v^2)\mathcal{R}_1(L, v)|_{v=c}. \quad (7.4)$$

Here $\mathcal{R}_1(L, v)$ is the right side of (6.1) with $\ell = (\log v - L)/2$, and L is held fixed.

Proof. The cumulant operator for marking by u is $\mathcal{D} = u\partial_u = v\partial_v$. Consequently the mean and variance are $\mathcal{D} \log f_n|_{u=1}$ and $\mathcal{D}^2 \log f_n|_{u=1}$. The logarithm of the leading term is

$$\log L_n(u) = 2\sqrt{vn} + \left(\frac{v}{2} - \frac{3}{4}\right) \log n + u(cA - 1) + \left(\frac{1}{4} - \frac{v}{2}\right) \log v - \log(2\sqrt{\pi}). \quad (7.5)$$

Differentiating this explicit holomorphic function and using $\log c = -\gamma$ gives (7.2)–(7.3). Differentiating the next term in (7.1) gives (7.4); its residual is controlled as just explained. $\square$

An entirely explicit form of these first refined polynomials is also useful. Write $q = Q(\ell_c)$, $q' = \ell_c + D$, and $\ell_c = (\log c - L)/2$. Then

$$M_1(L) = c^{3/2} \left(\frac{3}{2}q + \frac{1}{2}q' \right) - \frac{3}{8}c^{3/2} + \frac{1}{4}c^{1/2} + \frac{3}{32}c^{-1/2}, \quad (7.6)$$

$$V_1(L) = c^{3/2} \left(\frac{9}{4}q + \frac{3}{2}q' + \frac{1}{4} \right) - \frac{9}{16}c^{3/2} + \frac{1}{8}c^{1/2} - \frac{3}{64}c^{-1/2}. \quad (7.7)$$

These follow by writing the nonlogarithmic term in (6.1) as $-v^{3/2}/4 + v^{1/2}/2 - 3v^{-1/2}/16$ and using $v\partial_v \ell = 1/2$.

In particular,

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}K_n}{\sqrt{n}} = e^{-\gamma/2} \approx 0.749306001288449. \quad (7.8)$$

The decimal is uncertified. Riedel's numerical estimate 0.7719351399 [4, p. 3] is a finite-data estimate of the constant, accompanied there by caution about the fitted exponent. The positive $\frac{c}{2} \log n$ term explains a slow approach to the limit. The exact square-root limit belongs to the pre-existing expansive-assembly regime; the displayed lower-order corrections are specific to these component weights.

Corollary 7.2 (Credited Gaussian regime). *With $\sigma_n^2 = \sqrt{cn}/2$,*

$$\frac{K_n - \mathbb{E}K_n}{\sigma_n} \implies N(0, 1). \quad (7.9)$$

The same statement holds with σ_n^2 replaced by $\text{Var } K_n$.

Proof. Use the analytic logarithm of $f_n(e^t)/f_n(1)$ in a fixed complex neighborhood of $t = 0$. By (7.1) and Cauchy's estimates, its third derivative is $O(\sqrt{n} + \log n)$ there. Its second derivative at 0 is $\sigma_n^2 + O(\log n)$. For fixed real θ , substitute $t = i\theta/\sigma_n$ and subtract $t\mathbb{E}K_n$. Taylor's theorem gives

$$\log \mathbb{E} \exp \left(i\theta \frac{K_n - \mathbb{E}K_n}{\sigma_n} \right) = -\frac{\theta^2}{2} + o(1),$$

with cubic error $O(n^{-1/4})$ and second-derivative replacement error $O(n^{-1/2} \log n)$. The continuity theorem for characteristic functions proves (7.9). Equation (7.3) and Slutsky's theorem justify the variance normalization. $\square$

The proof recovers a known generic limit-law phenomenon with the normalization needed here. It is not a new universality assertion. No local limit theorem or quantitative distribution-distance bound is inferred from this fixed-frequency argument.

Remark 7.3 ([write] Riedel's conjecture, and what his fitted constants missed, 6 October 2026). *The source.* Riedel's note [4] has a paragraph "The conjecture" on p. 3 of 6, the same in the revised version (page footers dated 24.01.26) and in the original linked from A392471 (footers dated 13.01.26). With X the number of components of a uniformly random nested cycle partition on n nodes, and $\mathbb{E}[X]$ computed from his recurrences, the boxed conjecture is

$$\mathbb{E}[X] \sim N\sqrt{n},$$

where the constant N is left to be determined (readers able to contribute are asked to contact him). He adds that the asymptotics might instead involve a power n^M with M "close to but not equal to $1/2$ ", and reports that the numerical data suggest $M \geq 0.4704567259$ and $N \approx 0.7719351399$. The note gives neither the range of n nor the fitting method. Neither OEIS entry contains the conjecture (Section 1.1). His X is K_n : his EGF is $Q(z, u) = \exp(u \sum_{k \geq 1} Q_k(z))$ with $\sum_{k \geq 1} Q_k(z) = P(z) - 1$ (his p. 2), which is $F(z, u)$ of (1.2).

(a) *What is proved.* By (7.8) the boxed conjecture holds, with $N = e^{-\gamma/2} = 0.7493060\dots$; this answers his question for N . The alternative is excluded: for $N > 0$, $\mathbb{E}K_n \sim Nn^M$ holds only for $M = 1/2$, since $\mathbb{E}K_n/n^M = (\mathbb{E}K_n/\sqrt{n})n^{1/2-M}$ tends to $+\infty$ for $M < 1/2$ and to 0 for $M > 1/2$. The inequality $M \geq 0.4704567259$, as printed, is satisfied by $M = 1/2$; the value $N \approx 0.7719351399$ is not the constant: it exceeds $e^{-\gamma/2}$ by 0.0226. The source proves all of this in Theorem 7.1 but calls Riedel's number a finite-data estimate and does not say that the conjecture is settled; it is.

(b) *What the fits missed* [write]. Put $E_n = \mathbb{E}K_n$ and $m_0 = -0.977132\dots$ as in (7.2). Then

$$\frac{E_n}{\sqrt{n}} = e^{-\gamma/2} + \frac{\frac{c}{2} \log n + m_0}{\sqrt{n}} + O\left(\frac{(1 + \log n)^2}{n}\right),$$

and the numerator $\frac{c}{2} \log n + m_0$ is positive exactly when $n > e^{-2m_0/c} = 32.48\dots$; so $E_n/\sqrt{n} > e^{-\gamma/2}$ for all large n and approaches it only at the rate $(\log n)/\sqrt{n}$. For every fixed integer $\varrho \geq 2$ the two-point exponent satisfies

$$M_\varrho(n) := \frac{\log(E_{\varrho n}/E_n)}{\log \varrho} = \frac{1}{2} - (1 - \varrho^{-1/2}) \frac{c \log n}{2\sqrt{cn} \log \varrho} + O(n^{-1/2}),$$

so it lies below $1/2$ for all large n and tends to $1/2$ at the same slow rate. *Proof.* By (7.2), $E_n = \sqrt{cn}(1 + \delta_n)$ with $\delta_n = (\frac{c}{2} \log n + m_0)/\sqrt{cn} + O(n^{-1}(1 + \log n)^2) \rightarrow 0$. Since $\log(1 + x) = x + O(x^2)$, $\log(E_{\varrho n}/E_n) = \frac{1}{2} \log \varrho + \delta_{\varrho n} - \delta_n + O(n^{-1}(1 + \log n)^2)$, and

$$\begin{aligned} \delta_{\varrho n} - \delta_n &= \frac{\varrho^{-1/2}(\frac{c}{2} \log n + \frac{c}{2} \log \varrho + m_0) - (\frac{c}{2} \log n + m_0)}{\sqrt{cn}} + O\left(\frac{(1 + \log n)^2}{n}\right) \\ &= -(1 - \varrho^{-1/2}) \frac{c \log n}{2\sqrt{cn}} + O(n^{-1/2}). \end{aligned}$$

Dividing by $\log \varrho$ gives the claim. $\square$ A fitted exponent below $1/2$ and a fitted constant above $e^{-\gamma/2}$ are therefore what the proved law produces at every large finite n : the additive correction $\frac{c}{2} \log n + m_0$ in (7.2), which is not a power, is read by a fit as a change of exponent and of constant. Values (double-precision evaluation of E_n from recurrences with positive terms, agreeing with the exact rational E_{100} and with a 256-bit fixed-point evaluation at $n = 4500$):

n	100	1000	4500	10000	20000
$E_n/\sqrt{n}$	0.821954	0.786471	0.771935	0.766422	0.762634
$M_2(n)$	0.47781	0.48628	0.49089	0.49285	0.49428

For $1 \leq n \leq 40000$ every computed ratio $E_n/\sqrt{n}$ exceeds $e^{-\gamma/2}$, and the ratios decrease from $n = 4$ on.

(c) *Where Riedel's numbers may come from.* $E_{4500}/\sqrt{4500} = 0.77193514015$ agrees with his $N \approx 0.7719351399$ to 2.5×10^{-10} , and the ratio first falls below his value at $n = 4501$. This suggests, but the note does not say, that his N is the finite ratio at $n = 4500$. The write could not identify the computation behind $M \geq 0.4704567259$: the one-step log-log slopes $\log(E_n/E_{n-1})/\log(n/(n-1))$ for $n \leq 40000$, the two-point exponents $\log(E_{4500}/E_m)/\log(4500/m)$ for $m < 4500$, and least-squares fits of $\log E_n$ against $\log n$ over the windows $[a, 4500]$, $1 \leq a < 60$, and a few shorter windows were tried, and none reproduces it to its ten digits. Recorded only: nothing was submitted to the OEIS and Riedel was not contacted.

8 Asymptotic inversion with safe integer rounding

For sufficiently large real y , define

$$N(y) = \min\{n \geq 0 : a_n \geq y\}.$$

The sequence is strictly increasing for $n \geq 1$. Adding label $n + 1$ as a new singleton component is an injection; a single cycle on all $n + 1$ labels is outside its image. The equality $a_0 = a_1 = 1$ causes no problem for large y .

Put $\beta = c/2 - 3/4$. For fixed $J \geq 0$, define at sufficiently large real x

$$\begin{aligned} G_J(x) &= \log \Gamma(x + 1) + 2\sqrt{cx} + \beta \log x + \log C \\ &\quad + \log \left(1 + \sum_{j=1}^J x^{-j/2} R_j(\log x; 1) \right). \end{aligned} \tag{8.1}$$

The last logarithm is real and well defined eventually, and $G'_J(x) \sim \log x$. At integer n , Theorem 2.1 implies

$$\log a_n = G_J(n) + O(n^{-q}(\log n)^p), \quad q = (J + 1)/2, \quad p = 2J + 2. \tag{8.2}$$

For large y , let x_J be the unique sufficiently large real solution of $G_J(x_J) = \log y$.

Theorem 8.1 (Two-ceiling enclosure). *For each fixed J there exist $M_J > 0$ and a threshold such that, with*

$$\epsilon_J = M_J x_J^{-q} (\log x_J)^{p-1},$$

one has

$$\lceil x_J - \epsilon_J \rceil \leq N(y) \leq \lceil x_J + \epsilon_J \rceil. \quad (8.3)$$

If $\text{dist}(x_J, \mathbb{Z}) > \epsilon_J$, the two ceilings coincide and $N(y) = \lceil x_J \rceil$. No explicit certified M_J or onset threshold is supplied.

Proof. Bound the integer error in (8.2) by $Kn^{-q}(\log n)^p$. For $x = x_J$ sufficiently large, $\epsilon_J < 1$, $G'_J \geq (\log x)/2$ on $[x-2, x+2]$, and the error at integers in that interval is at most $2Kx^{-q}(\log x)^p$. Choose $M_J > 4K$. Then $n_+ = \lceil x + \epsilon_J \rceil$ satisfies

$$G_J(n_+) - G_J(x) \geq \frac{\log x}{2} \epsilon_J > 2Kx^{-q}(\log x)^p,$$

so $\log a_{n_+} > \log y$. Similarly $n_- = \lceil x - \epsilon_J \rceil - 1 < x - \epsilon_J$ satisfies $\log a_{n_-} < \log y$. Monotonicity gives (8.3). Enlarging the threshold ensures all comparisons used above. $\square$

A single ceiling of a finite continuous asymptotic expansion is not valid unconditionally: a target may lie arbitrarily close to an exact integer threshold. One must use the two-ceiling enclosure, prove the stated separation, or compare exact counts.

For a convenient starting approximation let $L = \log y$ and use the positive real branch of Lambert's W function:

$$x_0^* = \frac{L}{W(L/e)}. \quad (8.4)$$

It solves $x(\log x - 1) = L$, and $N(y) \sim x_0^*$. With $\alpha = c/2 - 1/4$ and $K_0 = \log(C\sqrt{2\pi})$, Stirling's formula gives

$$\begin{aligned} x_* &= x_0^* - \frac{2\sqrt{cx_0^*} + \alpha \log x_0^* + K_0}{\log x_0^*}, \\ x_J &= x_* + O((\log x_0^*)^{-2}). \end{aligned} \quad (8.5)$$

For justification, the displacement from x_0^* is $O(\sqrt{x_0^*}/\log x_0^*)$. Substituting (8.5) into $G_J - L$, the uncanceled derivative correction is $O(1/\log x_0^*)$; the quadratic curvature term is $O((\log x_0^*)^{-2})$ and the omitted Stirling and fixed-order corrections are smaller. Dividing by $G'_J \sim \log x_0^*$ proves the displayed error. Newton iteration or formal reversion of (8.1) gives further continuous orders. Every integer conclusion still needs (8.3) or an exact comparison. The companion's bounded exact inverse compares integer counts directly and does not use an uncertified asymptotic rounding rule.

Remark 8.2 ([write] The inversion against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`; its letters κ , d , v , X , L are written here with the subscript “vol” where they would collide with this report's.

- (a) *The starting value x_0^* of (8.4) is an instance, verbatim, of `p0:prop:factorial-core` with $(\kappa_{\text{vol}}, d_{\text{vol}}, L_{\text{vol}}) := (1, -1, L)$, $L = \log y > 0$: the phase is $X(\log X - 1)$, the solution is $v_{\text{vol}} = W_0(L/e)$ and $X_{\text{vol}} = e^{v_{\text{vol}}+1} = L/v_{\text{vol}} = x_0^*$ (using $v_{\text{vol}}e^{v_{\text{vol}}} = L/e$), and the side condition $\log X_{\text{vol}} - 1 > 0$ holds because $v_{\text{vol}} > 0$. The “positive real branch” of the source is W_0 .*

- (b) $N(y)$ is the staircase $N_*(y)$ of `p0:def:three-inverses` for the sequence (a_n) with $n_1 = 1$, for every $y > 1$: the sequence is strictly increasing for $n \geq 1$ (the injection argument at the start of this section), and for $y > 1 = a_0 = a_1$ the minima over $n \geq 0$ and over $n \geq 1$ agree.
- (c) *Theorem 8.1 is not an instance of `p0:thm:staircase`*, only an analogue of its item (2). Its proof compares the model G_J with $\log a_n$ at the integers and then uses monotonicity directly; no admissible interpolation of (a_n) appears, and x_J is the root of the model, not the inverse $\nu_{\mathcal{A}}(y)$ of an interpolation, so the hypothesis $|x_* - \nu(y)| \leq \eta$ of item (2) is not established. The conclusion has the two-ceiling form of item (2), and the paragraph after the proof describes the failure mode stated there. The Newton correction (8.5) is proved directly; no comparison with the volume's perturbation series is claimed.

No novelty is claimed for these inversions.

9 Exact recurrences and reproducible checks

The infinite products in the definitions reduce to finite exact computations at every fixed size. The companion uses a deliberately bounded range, not an unbounded enumeration promise.

Let $B_n^{(m)}$ count labeled permutations of n labels having at most one cycle of each length in $\{1, \dots, m\}$. With $B_0^{(0)} = 1$ and all other entries zero,

$$B_n^{(m)} = B_n^{(m-1)} + \begin{cases} \binom{n}{m} (m-1)! B_{n-m}^{(m-1)}, & n \geq m, \\ 0, & n < m. \end{cases} \quad (9.1)$$

The cycle's labels are chosen first and its cyclic order next; using the previous row enforces at most one such cycle. Thus $B_n = B_n^{(n)}$. The component containing label 1 gives the partial Bell recurrence

$$T(n, k) = \sum_{m=1}^n \binom{n-1}{m-1} B_m T(n-m, k-1), \quad n, k \geq 1. \quad (9.2)$$

All arithmetic in these recurrences is integer arithmetic. Their first five nonempty rows and row sums are

n	$T(n, 1), \dots, T(n, n)$	a_n
1	1	1
2	1, 1	2
3	5, 3, 1	9
4	14, 23, 6, 1	44
5	74, 120, 65, 10, 1	270

An independent rational-series check follows from

$$\log P(z) = \sum_{r \geq 1} \ell_r z^r, \quad \ell_r = \sum_{m|r} \frac{(-1)^{r/m+1}}{(r/m) m^{r/m}}.$$

Formal differentiation gives

$$np_n = \sum_{r=1}^n r \ell_r p_{n-r}, \quad nf_n(1) = \sum_{r=1}^n r p_r f_{n-r}(1), \quad p_0 = f_0(1) = 1. \quad (9.3)$$

A separate power-series multiplication checks $T(n, k) = n![z^n](P - 1)^k/k!$ directly. At small sizes, literal permutation cycle decomposition followed by enumeration of admissible groupings of cycles gives an additional combinatorial check.

The frozen OEIS fixtures contain the displayed numeric terms with public source URLs and retrieval metadata: 23 values of A007838, 22 values of A308338, and 55 entries of A392471. They are independent expected values, not values generated during a verification run. The companion checks those prefixes, complete counts and triangle through $n = 100$, independent rational count recurrences through $n = 100$, power-series triangle identities through $n = 30$, and literal enumeration through $n = 7$. Exact rational mean and variance samples come directly from the triangle. Optional SymPy checks verify the finite symbolic correction identities. See the package README for the commands and dependency versions used.

Evidence levels. The fixtures, finite recurrences, exact integer comparisons, rational moments, and symbolic identities are exact. They are finite checks and do not prove an asymptotic theorem. The fixed-order remainder, complex uniformity, all-circle suppression, moment expansion, CLT, and inverse enclosure are proved in the text using the credited component estimate. Their implied constants and onset thresholds have not been made effective here. All decimal constants or other numerical asymptotic diagnostics are uncertified; high arithmetic precision alone would not certify their errors. The package supplies no interval enclosure for the transcendental constant C and no empirical fitted constant as a substitute for its exact definition.

[write] **The package here, and the reruns, 6 October 2026.** “The companion” and “the package README” above are the delivered ones. In this report the companion programs are `code/companion-exact_nested.py`, `code/companion-verify.py`, `code/companion-test_exact_nested.py` and `code/companion-symbolic_checks.py` (delivered in `companion/`); the frozen fixtures and outputs are in `data/`; the companion’s README and provenance note are `companion-README.md` and `companion-PROVENANCE.md`; the release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`; `SOURCE_PROVENANCE.json` and `verification_receipt.json` are in `data/`; the checksum manifest is not shipped. `verify.py` reads `../data/published_fixtures.json` relative to its own folder, so the programs run only on a copy with the delivered names (the report README gives two routes). On such copies at the write (Windows, Python 3.14.4, SymPy 1.14.0): from a fresh extraction of the archive, `verify.py`, `verify.py -include-data`, `test_exact_nested.py` and `symbolic_checks.py` reproduce `full_verification.json`, `exact_data_100.json`, `tests.normal.json` and `symbolic.normal.json` once Windows line endings are converted; from the shipped files, the verifier and the regression tests reproduce their outputs the same way. The release tests need POSIX descriptor support and were not run.

10 Source coverage and attribution

The source review covered the three OEIS entries and Riedel’s revised January 2026 note, the 2006 hybrid asymptotics paper, and the later Erlihson–Granovsky expansive-assembly paper. It also checked the connection between the stated component weights and the hypotheses of that general assembly theory. This is bounded source coverage, not a worldwide historical-priority certificate.

The attribution is deliberately separated by mathematical role. Gutkovskiy’s 2019 OEIS entry supplies A308338’s defining EGF. Riedel supplies the nested-cycle interpretation and A392471. Greene and Knuth are credited for (1.4) by the inspected hybrid paper; the original book was not independently inspected. The hybrid paper itself develops the full component expansion, with root-of-unity periodic coefficient terms. Those component results do not by themselves establish the marked outer-exponential expansion proved here.

The inspected Erlihson–Granovsky preprint is the 19 June 2007 arXiv version of the paper

published in October 2008. Theorem and corollary numbers agree with the journal version, while some equation numbers differ. The reference to the 2004 total-component CLT is attribution verified in the later work, not a direct reading of the 2004 paper. That original paper was not inspected for this report.

The variance in (7.3) is derived independently from (7.5). There is a normalization inconsistency in the later source's displayed general- C covariance formula (preprint (4.54), published (4.18)): with its definition $b_r(u) = C\Gamma(r+1, u)$, the printed conditioning denominator is $\Gamma(p+2)$. The hypotheses allow every $C > 0$. At the interior choice $p = 1$, $C = 3$, $u = 1$, literal substitution gives $3/e - 18/e^2 < 0$, which cannot be a variance. This display is therefore not used to certify our constant. The paper remains the cited source for the general regime and its explicit historical attribution.

Within the checked sources, the explicit count prefactor, fixed-order complex-marked outer expansion, logarithmic moment refinements, and rounding-qualified inverse are the sequence-specific development of this report. No claim is made that no other work contains overlapping results. There is no external publication or amendment of an OEIS entry associated with this document.

Remark 10.1 ([write] The covariance display of Erlihson and Granovsky, corrected, 6 October 2026). In this remark $p, C, u, q, N, r_N, \delta, f_j, b_r, \nu, S_j, Z_N$ are the letters of [6]: the exponent and constant in $a_k \sim Ck^{p-1}$ ($k \rightarrow \infty$; every $p > 0$, $C > 0$ is allowed), a scaled component size, the number of strata, the total size, the scaling $r_N = h^{-1}N^{1/(p+1)}$ with $h = (C\Gamma(p+1))^{1/(p+1)}$, $\delta = 1/r_N$, and so on; they are not p_m , the amplitude C , the marking variable u or the q of Section 8.

The display. With $b_r(u) = C\Gamma(r+1, u)$ (preprint (4.53), published (4.17)), Theorem 4.3(i) states that the scaled counts $(\hat{\nu}_1, \dots, \hat{\nu}_q)$ of components of size at least $u_j r_N$ converge to a centred Gaussian vector with covariance

$$e_{mk} = b_{p-1}(u_s) - \frac{b_p(u_m)b_p(u_k)}{\Gamma(p+2)}, \quad s = \max(k, m)$$

(preprint (4.54); published (4.18) on p. 927, identical, read by the write), and Corollary 4.5 ($q = 1$, $u_1 = u$) gives $\hat{\nu}(u) \Rightarrow N(0, e_{11}(u))$ “for all $u \geq 0$ ”, the case $u = 0$ being the total number of components. The scaling of $\hat{\nu}_j$ (preprint (4.45), published (4.9)) multiplies $\nu_j - S_j$ by $(f_0(p-1)/S_0)^{1/2} \sim r_N^{-p/2}$.

(a) *The display fails.* At $p = 1$, $C = 3$ (for instance $a_k = 3$ for all k), $q = 1$, $u_1 = 1$: $b_0(1) = 3\Gamma(1, 1) = 3/e$, $b_1(1) = 3\Gamma(2, 1) = 6/e$ and $\Gamma(3) = 2$, so

$$e_{11} = \frac{3}{e} - \frac{36}{2e^2} = \frac{3}{e} - \frac{18}{e^2} = -1.33240\dots < 0,$$

which is not the variance of any random variable. More generally, at $u = 0$ the display gives $e_{11}(0) = C\Gamma(p) - C^2\Gamma(p+1)^2/\Gamma(p+2) = C\Gamma(p)(1 - Cp/(p+1))$, negative whenever $C > (p+1)/p$. The source found the first value (Section 10).

(b) *The correction.* The denominator should be $C\Gamma(p+2) = b_{p+1}(0)$:

$$e_{mk}^{\text{corr}} = b_{p-1}(u_s) - \frac{b_p(u_m)b_p(u_k)}{C\Gamma(p+2)},$$

which coincides with the printed display exactly when $C = 1$. This is what the conditioning computation gives. In the Boltzmann model at $\delta = 1/r_N$ the counts n_k are independent Poisson variables of means $a_k e^{-\delta k}$; with $\nu_j = \sum_{k \geq u_j r_N} n_k$ and the total mass $Z = \sum_k k n_k$, Riemann sums

give $r_N^{-p} \text{Cov}(\nu_m, \nu_k) \rightarrow C\Gamma(p, u_s) = b_{p-1}(u_s)$, $r_N^{-p-1} \text{Cov}(\nu_m, Z) \rightarrow C\Gamma(p+1, u_m) = b_p(u_m)$ and $r_N^{-p-2} \text{Var } Z \rightarrow C\Gamma(p+2) = b_{p+1}(0)$, and the covariance of a Gaussian vector conditioned on Z is $\text{Cov}(\nu_m, \nu_k) - \text{Cov}(\nu_m, Z) \text{Cov}(\nu_k, Z) / \text{Var } Z$. The corrected display is a variance: $e_{11}^{\text{corr}}(u) = C(\Gamma(p, u) - \Gamma(p+1, u)^2 / \Gamma(p+2)) > 0$, because the Cauchy–Schwarz inequality for $x^{(p-1)/2} e^{-x/2}$ and $x^{(p+1)/2} e^{-x/2}$ on (u, ∞) gives $\Gamma(p+1, u)^2 < \Gamma(p, u) \Gamma(p+2, u) \leq \Gamma(p, u) \Gamma(p+2)$. In the example of (a) it is $3/e - 6/e^2 = 0.29163\dots$. The write checked these limits and the conditioning formula, not the remaining steps of Section 5 of [6].

(c) *Where the factor C is lost.* The published (5.62) on p. 936 (preprint (5.121)) reads $\text{Var } Z_N = \sum_k k^2 a_k e^{-\delta k} \sim \Gamma(p+2) \delta^{-(p+2)}$; under $a_k \sim Ck^{p-1}$ the right side is $C\Gamma(p+2) \delta^{-(p+2)}$. The same $\Gamma(p+2)$ without C enters T_q in (5.69) (preprint (5.128)), while the proof of (5.72) on p. 938 invokes $\sum_j f_j(p+1) = C\Gamma(p+2)$. The covariance (4.13) of Theorem 4.1(ii) (preprint (4.49); published p. 926) carries the same denominator and fails in the same way: at $q=1$, $p=1$, $C=3$, $u_1=10$ its entry $f_0(0) - f_0(1)^2 / \Gamma(3)$ equals $-1.4956\dots$ (corrected: $1.5014\dots$). The write did not trace which step first drops C .

(d) *The display against this report.* The model here has weights $p_m \rightarrow c$ (Section 1), that is $p=1$ and $C=c$ in the letters of [6], with $N=n$ and $r_N = (n/c)^{1/2}$. Read at $u=0$, as their Corollary 4.5 is read there, with the normalization $r_N^{-1/2}$ (the limit of their scaling factor for $u_1 > 0$), the printed display gives $(K_n - S)/r_N^{1/2} \Rightarrow N(0, c(1 - c/2)) = N(0, 0.40384\dots)$ for a deterministic centring S . Corollary 7.2 and (7.3), proved here without their display, give $(K_n - \mathbb{E}K_n)/r_N^{1/2} \Rightarrow N(0, c/2) = N(0, 0.28072\dots)$, because $\text{Var } K_n/r_N \rightarrow c/2$. The two normalized variables differ by a deterministic shift t_n , and if $Y_n \Rightarrow N(0, \alpha)$ and $Y_n + t_n \Rightarrow N(0, \alpha')$ then (t_n) is bounded and each of its limit points t gives $N(t, \alpha) = N(0, \alpha')$, hence $\alpha = \alpha'$. So the printed display contradicts Corollary 7.2 in this model, and the corrected one, $e_{11}^{\text{corr}}(0) = c - c^2/(2c) = c/2$, agrees with it.

The source used the display for nothing and derived (7.3) independently; no report in the repository uses it. Recorded only: nothing was sent to the authors.

11 Further questions

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question. Questions 1–5 are the source’s, as delivered; what each lacks is what it names. Question 1: explicit constants in (1.4), in the off-saddle bound (4.4) and in Lemma 5.1. Question 2: control of $f_n(u)$ for u beyond a small disc about 1. Question 3: a global analysis of the secondary arcs beyond the all-arc bound (4.1). Question 4: complexity bounds and certified simplification for the calculator. Question 5: a faster exact algorithm with an independent check. The source’s non-claims are collected in Section 1.1. The intake found no false claim in the source and the write found none either; the two corrections concern outside literature (Remarks 7.3 and 10.1). Question 1 would also give the onset in Remark 7.3(b): $\mathbb{E}K_n/\sqrt{n}$ exceeds $e^{-\gamma/2}$ for all large n , and for every computed $n \leq 40000$, but no explicit n_0 is proved. Question 6 is the write’s.

Several extensions require additional work beyond the proved fixed-order statements.

1. Can the component estimates and Gaussian remainders be made effective enough to give certified constants and usable onset thresholds for the count equivalent and inverse enclosure?
2. Can complex marking be controlled beyond a small disc to prove a lattice local limit theorem, moderate deviations, or a quantitative normal-approximation bound for K_n ?

3. How do the periodic component corrections at secondary roots of unity appear in exponentially small outer-coefficient terms? A complete beyond-all-orders expansion would require a separate global analysis and is not implied by the present all-arc bound.
4. Can the fixed-order coefficient calculator be extended to higher corrections with explicit complexity bounds and certified symbolic simplification?
5. Can the exact Bell triangle be computed faster at large n while retaining an independent check and exact integer arithmetic?
6. [write] Remark 10.1 corrects the covariance displays (4.13) and (4.18) of Erlihson and Granovsky [6] by restoring the factor C in the variance of the total mass, and shows that the printed displays fail. Does their proof, with (5.62) and (5.69) corrected, give exactly the corrected displays, and does any other statement of their Sections 4 and 5 depend on the missing factor? What is missing is a line-by-line audit of their Section 5; the write checked only the conditioning computation and the three limits it uses.

These are research questions, not consequences of the finite numerical or symbolic checks.

References

- [1] The OEIS Foundation, *A007838*, permutations with distinct cycle lengths. <https://oeis.org/A007838>. Entry inspected 3 October 2026.
- [2] I. Gutkovskiy, *A308338*, OEIS entry dated 20 May 2019. <https://oeis.org/A308338>. Entry inspected 3 October 2026.
- [3] M. Riedel, *A392471*, triangle counting nested-cycle assemblies by component count. <https://oeis.org/A392471>. Entry inspected 3 October 2026.
- [4] M. Riedel, *Nested Cycle Partitions: a conjecture*, revised 24 January 2026 author-hosted note, 6 pp. <https://pnp.mathematik.uni-stuttgart.de/iadm/Riedel/papers/ncp-comp2.pdf>. Original 13 January version linked from OEIS: <https://oeis.org/A392471/a392471.pdf>. The nested-cycle specification and component-count discussion were inspected directly.
- [5] P. Flajolet, É. Fusy, X. Gourdon, D. Panario, and N. Pouyanne, *A Hybrid of Darboux’s Method and Singularity Analysis in Combinatorial Asymptotics*, Electronic Journal of Combinatorics **13**(1) (2006), R103, 35 pp. <https://arxiv.org/abs/math/0606370>. Section 3 and Proposition 1 of the 15 June 2006 version were inspected. The Greene–Knuth attribution is indirect through that work; the book itself was not inspected.
- [6] M. M. Erlihson and B. L. Granovsky, *Limit shapes of Gibbs distributions on the set of integer partitions: The expansive case*, Annales de l’Institut Henri Poincaré, Probabilités et Statistiques **44**(5) (2008), 915–945. <https://doi.org/10.1214/07-AIHP129>. Published text: <https://numdam.org/item/10.1214/07-AIHP129.pdf>. The assembly weights, expansive scaling, Theorems 4.1 and 4.3, and Corollary 4.5 were checked against the 2007 preprint; relevant published equation numbering and covariance text were also checked.
- [7] M. Erlihson and B. Granovsky, *Reversible coagulation-fragmentation processes and random combinatorial structures: asymptotics for the number of groups*, Random Structures & Algorithms

25 (2004), 227–245. Cited here solely through the explicit historical attribution in [6, Corollary 4.5 and reference 12]; this paper was not inspected directly.